MILITARY SPECIFICATION

NUT, SELF-LOCKING, SPLINE DRIVE, ASSEMBLED WASHER, NICKEL ALLOY 718, 220 ksi Ftu, 900°F, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-locking Nickel Alloy 718 corrosion resistant nuts for use in temperatures up to and including 900°F, in conjunction with expandable shank bolts conforming to MIL-B-85667.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-P-35 Passivation Treatments for Corrosion Resisting Steel

PPP-H-1581 Hardware (Fasteners and Related Items) Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 5311), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (continued)

MILITARY

MIL-H-6875	Heat Treatment of Steel (Aircraft Practice), Process for
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for
MIL-W-8982	Wrenches, Splined, High Strength, Thin Wall, General Specification for
MIL-L-46010	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting
MIL-B-85667	Bolt, Straight Shank, Expandable Sleeve, Protruding and Flush Head, General Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-410	Non-destructive Testing Personnel Qualification and Certification
MIL-STD-1312	Fastener Test Methods
MIL-STD-6866	Inspection, Liquid Penetrant
MS33588	Nut, Self-Locking, Aircraft, Design and Usage Requirements For
MS33787	Wrenching Element, External Spline, Dimensions for

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

AMS 5525	Steel, Sheet, Strip & Plate, Corrosion & Heat Resistant
AMS 5662	Bar, Forging, Rings, Nickel Base, 19CR, 19FE, 3.1MO, 5.1 (Cb+Ta) 0.9Tl, 0.5AL consumbale electrode or Vacuum Induction melted, solution heat treated

(Application for copies should be addressed to Society of Automotive Engineers, Inc. 400 Commonwealth Drive. Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern. If a specific requirement specified herein is not required for an item, it shall be so indicated on the specification sheet.

3.2 Qualification. The nut furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List at the time set for opening of bids, see 4.3 and 6.3.

3.3 Retention of qualification. To maintain status on a Qualified Products List, certification shall be submitted at two year intervals to indicate continued compliance with the requirements of this specification, see 4.3.4.

3.4 Definitions. For definitions of terms used in this specification, see 6.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

3.5 Materials.

3.5.1 Nut. The nut shall be fabricated from nickel alloy 718 as specified in the applicable specification sheet.

3.5.2 Washer. The washer shall be fabricated from corrosion-resistant steel (A-286) as specified in the applicable specification sheet.

3.6 Design. Nut design shall conform to the applicable specification sheet.

3.7 Construction. Nuts shall be of the prevailing torque-type. The locking device shall not operate by means of separate movement as a result of installation and shall not depend upon pressure on the bearing surface for locking action.

3.7.1 Wrenching element. Wrenching element shall be in accordance with MS33787.

3.7.2 Bearing surface. After removal of the washer from the nut, the bearing surface shall be normal to the axis of the pitch diameter of the threads within the values specified on the applicable standard or drawing.

3.7.3 Threads. Unless otherwise specified, thread dimensions shall be in accordance with MIL-S-8879. Threads used on the locking device may be displaced in any manner which will provide self-locking nuts conforming to this specification. Thread gaging shall be performed after plating in accordance with the applicable thread specification, (except that the nut with locking device incorporated but without lubricant shall allow the go plug gage members to enter a minimum of three-fourths turn before engagement of the locking device). When lubricant prevents use of standard gages, the nut shall permit free rotational (finger torqued) bolt engagement of at least three-fourths turn.

3.8 Surface treatment. The nut shall be passivated in accordance with QQ-P-35.

3.9 Dry film lubricant. The nut shall be lubricated with dry film lubricant conforming to MIL-L-46010.

3.10 Axial strength. The nuts shall have a minimum axial strength of not less than the values specified in table I, see 4.5.2.

3.11 Torque.

3.11.1 Wrench torque. The nuts shall withstand the wrench torque values specified in table I, when tested as specified in 4.5.3.1. Upon completion of the test, the nut shall show no indications of permanent deformation which may prevent its proper application and removal with a socket-type wrench conforming to MIL-W-8982. The nuts shall show no evidence of cracking, see 4.5.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

3.11.2 Torque effectivity, room temperature. The nuts shall not exceed the room temperature maximum locking torque nor produce less than the minimum breakaway torque specified in table III, see 4.5.3.2.

3.11.2.1 Torque effectivity, 900°F. The nuts shall not exceed the room temperature after bake maximum locking torque nor produce less than the minimum breakaway torque specified in table III, see 4.5.3.2.1.

3.11.3 Locking torque. The nuts shall not exceed the room temperature maximum locking torque nor produce less than the 1st cycle minimum breakaway torque specified in table III, see 4.5.3.3.

3.11.4 Permanent set. The nuts shall not exceed the maximum locking torque nor be less than the minimum breakaway torque specified in table III, see 4.5.3.4.

3.12 Accelerated vibration. The nuts shall withstand the vibration test specified in 4.5.4.

3.13 Stress durability. The nuts shall show no evidence of cracking, see 4.5.5.

TABLE I. Axial Strength and Wrench Torque

Nominal Size	Cross Sectional Area (Sq-In.)	Axial Strength Minimum (Lbs.)	Wrench Torque Minimum (In-Lbs.)
10-32	.0226	4,970	80
1/4-28	.0404	8,890	210
5/16-24	.0640	14,100	450
3/8-24	.0951	20,900	730
7/16-20	.1288	28,300	1,100
1/2-20	.1717	37,800	1,500
9/6-18	.2176	47,900	2,100
5/8-18	.2724	59,900	2,700
3/4-16	.3952	86,900	4,600
7/8-14	.5392	119,000	7,600
1 -12	.7027	155,000	10,400
1-1/8-12	.9007	198,000	13,500
1-1/4-12	1.1232	247,000	16,500
1-3/8-12	1.3703	301,000	18,300
1-1/2-12	1.6420	361,000	22,000

Axial Strength Values determined from Formula Wa = Ftu A;
Where A = Cross sectional area (Sq-In.) as tabulated.
Area based on maximum pitch diameter of bolt thread.
Ftu = 220 KSI Wa = Axial Strength (lbs.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

TABLE II. Stress Durability and Stress Corrosion

Nominal Size	Nominal Grip Length of Bolt (In.)	Bolt Elongation (In.)
10-32	2.00	.0070
1/4-28	2.00	.0073
5/16-24	2.00	.0076
3/8-24	2.00	.0080
7/16-20	2.50	.0096
1/2-20	2.50	.0101
9/16-18	2.50	.0103
5/18-18	2.75	.0114
3/4-16	2.75	.0122
7/8-14	2.75	.0126
1 -12	3.00	.0138
1-1/8-12	3.00	.0144
1-1/4-12	3.00	.0149
1-3/8-12	3.00	.0154
1-1/2-12	3.00	.0159

(a) Test bolt shall be stressed by torquing the nut until elongation is within 0.0005 of value specified. This elongation will indicate a stress of 50% of the minimum axial strength.

(b) Bolt elongations are calculated using bushing assembly lengths "A" (dimensions "L" per table IX).

(c) Other bolt grip lengths permissible. Calculate elongation per MIL-STD-1312-14, using values for "A" equal to nominal grip length plus three times the thread pitch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

TABLE III. Torque values.

Nominal Size	Seating Torque For Torque Effectivity Test (In-Lbs.)	Maximum Locking Torque 1/		Minimum Breakaway Torque (In-Lbs.)	Assembly Torque Vibration (In-Lbs.)
		Room Temperature (In-Lbs.)	Room Temperature After Bake (In-Lbs.)		
10-32	76	18	36	2.0	36
1/4-28	190	30	60	3.5	60
5/16-24	400	60	120	6.5	120
3/8-24	655	80	160	9.5	160
7/16-20	1,000	100	200	14.0	200
1/2-20	1,260	150	300	18.0	300
9/16-18	1,800	200	400	24.0	400
5/8-18	2,350	300	600	32.0	600
3/4-16	4,000	400	800	50.0	
7/8-14	6,500	600	1,200	70.0	
1 -12	9,300	800	1,600	90.0	
1-1/8-12	11,700	900	1,800	117.0	
1-1/4-12	14,400	1,000	2,000	143.0	
1-3/8-12	17,000	1,100	2,200	165.0	
1-1/2-12	19,800	1,250	2,500	195.0	

1/ Values apply to nuts as lubricated by manufacturer.

3.14 Stress corrosion. The nuts shall show no evidence of cracking when tested as specified in 4.5.6.

3.15 Discontinuities. Discontinuities in the nuts shall not exceed the depth limitations shown in table IV when examined as specified in 4.5.7. Care shall be exercised not to confuse cracks with discontinuties, see 6.4.3. Cracks are not permitted in any location.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

TABLE IV. Discontinuity depth limits, maximum, inch 1/.

Nominal Thread Size	Limits	Rejectable Criteria
10-32	.005	Any crack, regardless of location. Discontinuties in excess of specified limits.
1/4-28	.005	
5/16-24	.005	
3/8-24	.006	
7/16-20	.007	
1/2-20	.008	
9/16-18	.009	
5/8-18	.010	
3/4-16	.010	
7/8-14	.010	
1 -12	.011	
1-1/8-12	.011	
1-1/4-12	.012	
1-3/8-12	.012	
1-1/2-12	.012	

1/ Samples of nuts having indications may be sectioned and micro-examined to determine whether the indications are due to tool or die marks, or due to discontinuities, and to determine conformance of discontinuities to the depth limits specified in table IV.

3.16 Heat treatment. The nuts and washers shall be heat treated in accordance with MIL-H-6875 to develop the mechanical properties as specified herein.

3.17 Workmanship. Workmanship shall be consistent with the type or product, finish, and the class of thread fit specified. Sharp edges shall be broken; hanging burrs and slivers shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification Inspection (4.3)

b. Quality Conformance Inspection (4.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

4.3 Qualification inspections.

4.3.1 Sampling instructions. Qualification test samples (see table V) shall consist of 50 nuts for each size upon which qualification is desired. All bolts, screws, and mandrels necessary for tests specified herein shall be furnished by the manufacturer. Complete description and name of lubricating material used on the nuts shall be furnished. Samples shall be identified as required and forwarded to the activity responsible for qualification designated in the letter of authorization from that activity, see 6.3.

TABLE V. Qualification test plan.

Test	Paragraph	Nuts Required	Bolts or Studs Required
Examination of Product	4.5.1	All	0
Thread Fit	4.5.1	All	0
Bearing Surface Squareness	4.5.1.1	3	0
Axial Strength	4.5.2	5	5
Wrench Torque	4.5.3.1	5	5
Torque Effectivity, Room Temp. (Locking Element)	4.5.3.2	5	5
Torque Effectivity, 450°F (Locking Element)	4.5.3.2.1	5	5
Locking Torque	4.5.3.3	3	3
Permanent Set	4.5.3.4	3	3
Stress Corrosion	4.5.6	6	6
Accelerated Vibration	4.5.4	5	5
Stress Durability	4.5.5	6	6
Discontinuities	4.5.7	All	0

4.3.2 Inspections. The qualification inspections shall consist of all the tests and examinations specified in 4.5.

4.3.3 Certified test report. The manufacturer shall furnish a certified test report showing that the manufacturer's product satisfactorily conforms to this specification. The test report shall include, as a minimum, actual results of the tests specified herein. When this report is submitted, it shall be accompanied by a dated drawing which describes the manufacturer's product by specifying pertinent dimensions and tolerances, composition of material selected, coating and the heat treat. The manufacturer's part number for each size shall be included on the above drawing.

4.3.4 Retention of qualification. Certification shall be requested by the Naval Air Development Center (NAVAIRDEVCEN), Code 6013, Warminster, PA 18974, from each manufacturer. NAVAIRDEVCEN will forward certification to the Naval Air Engineering Center (NAVAIRENGCEN), Code 5311, who is acting as agent for

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

the Naval Air Systems Command (NAVAIR), AIR-53033G). Certification shall be at the time of the two year review and shall be signed by a responsible official of management, attesting that the listed product(s) is still available from the listed plant, can be produced under the same conditions as originally qualified; (i.e., same process, materials, construction, design, manufacturer's part number, or designation); and meets the requirements of the current issue of the specification. Failure to provide the certification will be cause for removal from the QPL. After completion of the certification review, the QPL will be reprinted to show the date of certification. (DD Form 1718, Certification of Qualified Products, shall be used for obtaining certification.)

4.4 Quality conformance inspections. The quality conformance inspections shall consist of those examinations and tests listed in 4.4.2 on samples selected in accordance with sampling plans A, B, and C.

4.4.1 Selection of samples. Sample nuts shall be selected at random from each inspection lot as specified herein.

4.4.1.1 Inspection lot. A lot shall consist of finished nuts of the same type and diameter fabricated by the same process, from one heat of material, heat treated in the same manner, and produced as one continuous run, order, or part thereof.

4.4.1.2 Sampling Plan A. Sampling sizes for examination of product and thread fit (4.5.1) and bearing surface squareness (4.5.1.1) shall be in accordance with MIL-STD-105, inspection level I, as specified in table VI. The acceptance and rejection criteria shall be applied to the following acceptance quality levels (AQLs) pertaining to the corresponding class of characteristics:

Major AA	0.0 percent
Major A	0.4 percent
Major B	1.5 percent
Major C	2.5 percent
Major D	4.0 percent

4.4.1.2.1 Classification of defects. The classification of defects for self-locking nuts shall be:

Major AA

Visual inspection of package marking.

Major A

Visual presence of locking configuration.
Bearing surface squareness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

Major B

Thread Fit.
Surface Plating.

Major C

Wrench Size - Height of wrenchable portion.
Overall Height - Diameter of bearng surface.
Loose or hanging burrs - Concentricity of threads to base diameter.
All dimensional characteristics not covered above.

Major D

Part identification per applicable standard or drawing.

TABLE VI. Inspection Sampling Plan A

Lot Size	0.4% AQL			1.5% AQL			2.5% AQL			4.0% AQL		
	Sample Size	Ac	Re	Sample Size	Ac	Re	Sample Size	Ac	Re	Sample Size	Ac	Re
Under 26	25	0	1	8	0	1	5	0	1	3	0	1
26 thru 50	32	0	1	8	0	1	5	0	1	13	1	2
51 thru 90	32	0	1	8	0	1	20	1	2	13	1	2
91 thru 150	32	0	1	32	1	2	20	1	2	20	2	3
151 thru 280	32	0	1	32	1	2	32	2	3	32	3	4
281 thru 500	32	0	1	50	2	3	50	3	4	50	5	6
501 thru 1,200	125	1	2	80	3	4	80	5	6	80	7	8
1,201 thru 3,200	125	1	2	125	5	6	125	7	8	125	10	11
3,201 thru 10,000	200	2	3	200	7	8	200	10	11	200	14	15
10,001 thru 35,000	200	2	3	315	10	11	315	14	15	315	21	22
Over 35,000	200	2	3	500	14	15	500	21	22	315	21	22

Where sample size is equal to or greater than lot size 100% inspect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

4.4.1.3 Sampling Plan B. For the axial strength test (4.5.2), Locking Torque Test (4.5.3.3), and Permanent Set (4.5.3.4), the samples shall be selected in accordance with table VII.

TABLE VII. Sampling Plan B

Lot Size	Sample Size	Acceptance Number
Under 10,000	5	0
10,000 to 50,000	10	0
50,001 to 100,000	15	0
Over 100,000	27	1

4.4.1.4 Sampling Plan C. For Stress Durability (4.5.5), sample size shall be 5 for each lot and the acceptance number zero.

4.4.2 Tests. The quality conformance tests shall consist of:

a. Examination of Product (4.5.1 and 4.5.1.1)

b. Axial Strength (4.5.2)

c. Locking Torque (4.5.3.3)

d. Discontinuities (4.5.7)

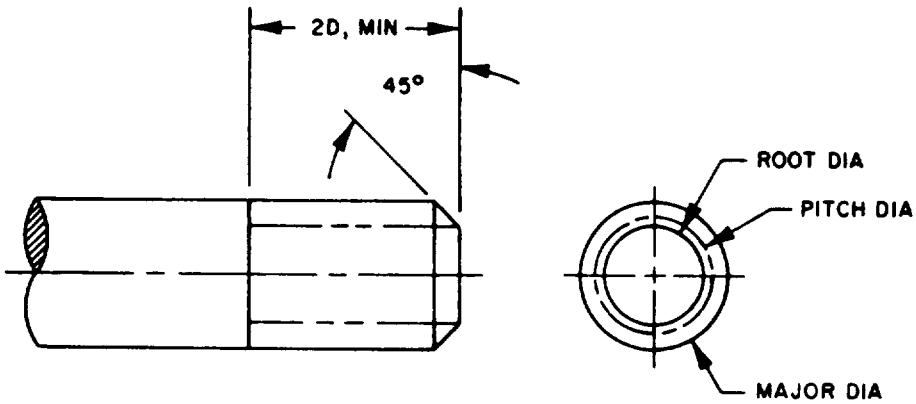
4.5 Test methods.

4.5.1 Examination of product. The nuts shall be examined for conformance to this specification and applicable approved standards with respect to material, workmanship, dimensions, design and construction, and finish.

4.5.1.1 Bearing surface. After removal of washer from the nut bearing surface values shall be measured by means of a table squareness gage, with a seating surface diameter equal to or greater than the base diameter of the nut being examined, in conjunction with a feeler gage, with a class 3A threaded member at engagement in the locking device, and turned finger tight.

4.5.2 Axial strength. The nuts shall be assembled on test bolts as specified in table VIII, having minimum grip length equal to three times the bolt diameter, and subjected to the axial strength load specified in table I. Test method and fixtures shall be in accordance with MIL-STD-1312-8. The nuts shall support the axial strength load without rupture. For qualification, apply load until nut or bolt failure occurs. For quality conformance, tests may be discontinued after minimum load has been achieved. Bolt failure below the rated axial strength of the nut due to defective bolt is not cause for rejection. In the event of such bolt failure, this test shall be repeated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.



DETAILS:

MATERIAL: STEEL

HARDNESS: HEAT TREATED TO ROCKWELL C60 to 64.

THREADS: THE THREAD FORM SHALL CONFORM TO THAT SHOWN IN MIL-S-8879. THE THREADS SHALL BE RIGHT HAND. THE THREAD DIMENSIONS SHALL BE AS SHOWN IN TABLE XI.

SURFACE ROUGHNESS: 20 MICROINCHES MAXIMUM IN ACCORDANCE WITH ANSI B46.1.

THE MANDREL END SHALL HAVE A 45-DEGREE CHAMFER EXTENDING BELOW THE ROOT DIAMETER. THE RESULTING SHARP FEATHER EDGE OF THE INCOMPLETE THREAD SHALL BE REMOVED BY STONING.

NOTE: LUBRICANT SHALL BE USED ONLY WHEN THE MANDREL IS USED WITH CORROSION-RESISTING STEEL NUTS.

FIGURE 1. Mandrel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

4.5.3 Torque.

4.5.3.1 Wrench torque. The nuts shall be installed on test bolts as specified in table VIII with a steel fixture in accordance with figure 2, and torqued to the wrench torque test values specified in table I. The nuts shall then be removed and reinstalled to this torque fourteen additional tmes. Wrenches shall be of the socket type conforming to MIL-W-8982. Deformation which interferes with the proper application and removal of the nut with the wrench is sufficient cause for rejection.

4.5.3.2 Torque effectivity-room temperature. The nuts shall be installed on test bolts as specified in table VIII with test fixtures in accordance with figure 2 and torqued to the seating torque values specified in table III. The nuts shall then be removed and reinstalled (seated) fourteen additional times. The maximum locking torque and minimum breakaway torque shall be that for the 1st, 7th and 15th cycles shall be recorded. The minimum breakaway torque shall be that torque required to start relative motion, between the nut and bolt, after the nut has been unseated and backed off one-half-turn.

4.5.3.2.1 Torque effectivity-after 900° bake. The nuts shall be installed on test bolts as specified in table VIII with test fixtures in accordance with figure 2 and torqued to the seating torque values specified in table III. The assemblies shall then be baked for one hour at 900°F (-0 +25°) and cooled to room temperature. The nuts shall then be removed and reinstalled (seated and baked) four additional times. The maximum locking torque and minimum breakaway torques for the 1st and 5th cycles shall be recorded. The minimum breakaway torque shall be that torque required to start relative motion, between the nut and bolt, after the nut has been unseated and backed off one-half turn.

4.5.3.3 Locking torque. The nuts shall be installed and removed on test bolts, as specified in table VIII, one time, recording maximum locking torque and minimum breakaway torque during the one cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

TABLE VIII. *Type of test bolt or stud.

Type	Test	Paragraph Number
I	Axial Strength, Room Temperature	4.5.2
I	Stress Durability	4.5.5
I	Torque Effectivity, Room Temperature (Locking Element)	4.5.3.2
I	Wrench Torque	4.5.3.1
I	Locking Torque	4.5.3.3
II	Permanent Set	4.5.3.4
I	Torque Effectivity, 450° (Locking Element)	4.5.3.2.1
I	Stress Corrosion	4.5.6
I	Accelerated Vibration	4.5.4

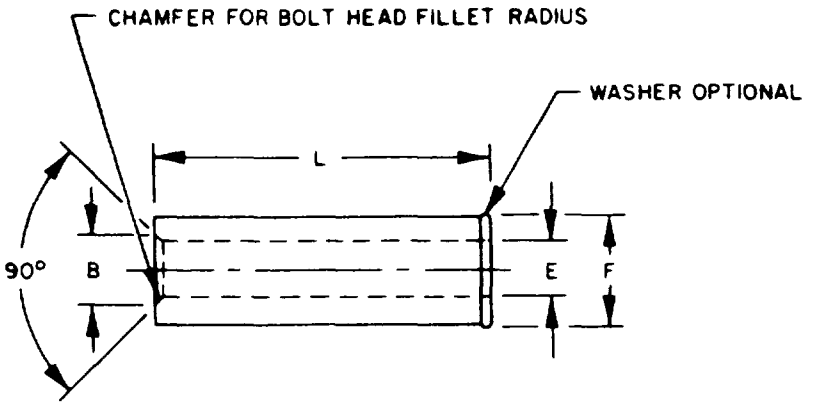
*Type I Rolled threads in accordance with MIL-S-8879 thread form, having 220 KSI (minimum) strength level. If bolts are used, head configuration is optional.

II A. Maximum Stud - 160 KSI (minimum) strength level with rolled threads in accordance with MIL-S-8879, except that the pitch diameter shall be established at 75%, +.0000-.0004 of the tolerance range of class 3A, above the minimum pitch diameter subject to check with functional pitch diameter tri-roll gages, see table XI.

B. Minimum Mandrel - In accordance with figure 1. Threads shall conform to table XI. It shall be checked with pitch diameter tri-roll gages.

Test bolts specified in this table are for referee purposes. Bolts or threaded studs may be used for routine testing. Threads may be rolled before or after heat treatment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.



Test Fixture for Torque Effectivity and Wrench Torque.

Material: Nickel Alloy 718 per AMS 5662

Heat Treat: 220 KSI Ftu

Ends of bushing and bushing plus washer to be flat and parallel within .002. Surface texture of ends: 32 microinches per ANSI B46.1.

FIGURE 2. Test fixture.

TABLE IX. Dimensions for torque test bushing assembly.

Nut Nominal Thread Size (Ref.)	B Diameter ± .005	E Diameter +.000 -.010	F Diameter Minimum	L ±.010 (a)
.1900-32	.285	.205	Diameter of Nut Base or Twice the Diameter of the Bolt Whichever is Greater	2.125
.2500-28	.345	.265		2.125
.3125-24	.407	.327		2.125
.3750-24	.502	.390		2.125
.4375-20	.564	.452		2.625
.5000-20	.627	.515		2.625
.5625-18	.689	.577		2.625
.6250-18	.783	.640		2.875
.7500-16	.908	.765		2.938
.8750-14	1.033	.890		2.938
1.0000-12	1.158	1.015		3.188
1.1250-12	1.283	1.140		3.188
1.2500-12	1.440	1.265		3.188
1.3750-12	1.565	1.390		3.188
1.5000-12	1.690	1.565		3.188

(a) Length "L" may include one washer of equal material and hardness. Length "L" may vary to accommodate other bolt lengths - see table II, note c.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

4.5.3.4 Permanent set. Permanent set shall be subjecting the nut at room ambient temperature, to one complete installation and removal cycle on a maximum stud in accordance with table VIII and then repeating the test cycle with the same clean nut on a minimum pitch mandrel, in accordance with table VIII. Three of the nuts submitted shall be subjected to this test.

4.5.4 Vibration. Sample nuts with bolts of the size and quantities specified in table X shall be vibrated in accordance with MIL-STD-1312-7. Vibration life, for each lot of five specimens tested in accordance with MIL-STD-1312-7, of less than 45,000 cycles shall be cause for rejection.

TABLE X. Vibration requirements.

Nut Size	Bolt Grip (Inch)	MIL-STD-1312-7 Bolts and Nuts Required (Min. of each) 900°F
10-32	1.125	10
1/4-28	1.188	10
5/16-24	1.188	10
3/8-24	1.188	10
7/16-20	1.188	10
1/2-20	1.188	10
9/16-18	1.188	10
5/8-18	1.188	10

4.5.4.1 Vibration tests on nuts larger than 5/8-inch size are waived, provided that 5/8-inch nuts and smaller of the same type and design of locking device have satisfactorily passed the vibration test.

4.5.4.2 Preparation for vibration test to MIL-STD-1312-7.

4.5.4.3 Accelerated vibration. The nuts shall be assembled in accordance with figure 2 of MIL-STD-1312-7, with bolts and torque values as specified in table III. The nuts shall then be removed and reinstalled to this torque four additional times before being vibrated.

4.5.4.4 Baking of test specimens. Half of the nuts shall be assembled on the appropriate bolts and baked for six hours at 450°F. The baked specimens shall be allowed to cool slowly in air to room temperature. The nuts shall then be removed and reinstalled to torque values specified in table III for four additional times before being vibrated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

4.5.4.5 Method. Use procedure of MIL-STD-1312-7.

4.5.4.5.1 Determination shall be made throughout the test to guarantee that assembly is traversing the entire length of the slots in the test fixture. The test shall be run for 45,000 cycles except that it shall be stopped prior to the completion of the 45,000 cycles in the event a nut becomes disassembled from the bolt. The nut samples shall be examined under 10X magnification for cracks.

4.5.4.5.2 The nuts shall be considered to have failed to pass the vibration test under the following conditions:

a. If any structural failure, such as broken segments, locking inserts falling out, or cracks occurring in the nuts during the test, provided failure is not the result of failure of the bolt.

b. If any nut comes completely off the bolt or can be turned completely on or off the bolt with the fingers during or after completion of 45,000 cycles.

c. If relative rotation between any nut and bolt exceeds 360 degrees.

4.5.5 Stress durability. Test in accordance with MIL-STD-1312-14, as specified in table VIII and torqued until the bolt elongates the amount specified in table II. The assemblies shall then be baked for 96 hours at 450°F (-0° + 25°) and cooled at room temperature. After removing nuts, polish bearing surface of nuts and examine at 10X to 50X magnification for cracks.

4.5.6 Stress corrosion. Test in accordance with MIL-STD-1312-9. The nuts shall be installed on test bolts as specified in table VIII and torqued until the bolt elongates the amount specified in table II. After 500 hours of alternate immersion, the parts shall be disassembled and the nut bearing surface shall be polished and examined, at 10X to 50X magnification, for cracks.

4.5.7 Cracks and discontinuities. Fluorescent penetrant inspection shall be used to determine the presence of cracks and discontinuities such as laps, seams and inclusions. Fluorescent penetrant indications of themselves shall not be cause for rejection. If indications are considered cause for rejection, representative samples shall be taken from these nuts showing indications and these samples shall be further examined by microexamination to determine whether the indicated discontinuities exceed the limits specified herein.

4.5.7.1 Fluorescent penetrant inspection. Fluorescent penetrant inspection shall be performed in accordance with MIL-STD-6866. Personnel conducting magnetic particle inspection shall be qualified in accordance with MIL-STD-410.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

TABLE XI. Thread dimensions for maximum stud and minimum mandrel for permanent set test.

Maximum 3A Stud			Minimum 3A Mandrel			
Size	Pitch Diameter Maximum	Pitch Diameter Minimum	+0.0000 -0.0004 Major Diameter	+0.0000 -0.0004 Pitch Diameter	Tolerance in lead (per inch)	Tolerance on half angle of thread (in minutes)
10-32	.1691	.1687	.1840	.1674	±.0003	±15
1/4-28	.2261	.2257	.2435	.2243	±.0003	±15
5/16-24	.2847	.2843	.3053	.2827	±.0003	±15
3/8-24	.3471	.3467	.3678	.3450	±.0003	±15
7/16-20	.4042	.4038	.4294	.4019	±.0003	±15
1/2-20	.4667	.4663	.4919	.4643	±.0003	±15
9/16-18	.5255	.5251	.5538	.5230	±.0003	±10
5/8-18	.5880	.5876	.6163	.5854	±.0003	±10
3/4-16	.7084	.7080	.7406	.7056	±.0003	±10
7/8-14	.8275	.8271	.8647	.8245	±.0003	±10
1 -12	.9448	.9444	.9886	.9415	±.0003	±10
1-1/8-12	1.0697	1.0693	1.1136	1.0664	±.0003	±10
1-1/4-12	1.1947	1.1943	1.2386	1.1913	±.0003	±10
1-3/8-12	1.3197	1.3193	1.3636	1.3162	±.0003	±10
1-1/2-12	1.4447	1.4443	1.4886	1.4411	±.0003	±10

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Self-locking nuts shall be preserved, packaged, packed, and marked for shipment in accordance with PPP-H-1581. Preservation and packaging shall be level A or C as specified in the contract or purchase order, see 6.2.

5.2 Packing shall be level A, B or C as specified in the contract or purchase order, see 6.2.

5.3 Marking of shipments. In addition to any special requirements of the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The nuts are intended for use in airframes and airborne mechanical systems in combination with high fatigue 220 KSI bolt applications where the maximum temperatures do not exceed 900°F.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Military part number, see 3.1.

c. Applicable levels of packaging and packing.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the Qualified Products Lists (QPL-85730) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Navy Department, Washington, Dc 20361, and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Qualification tests will be authorized only upon presentation of certified test reports indicating that the nuts conform to this specification, see 4.3.3 and the applicable military specification sheet or a standard approved by the activity responsible for qualification.

6.4 Definitions applicable to the tests under this specification.

6.4.1 Installed. A nut is considered installed when a minimum of two thread pitches plus the chamfer of the male thread extend beyond the top of the nut.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

6.4.2 Removal cycle. The removal cycle is considered complete when the locking device is disengaged.

6.4.3 Crack. A crack is a clean break passing through the grain or grain boundary without inclusion of foreign elements.

6.4.4 Minimum breakaway torque. The minimum breakaway torque is that torque required to start nut rotation from a fixed position during a removal cycle with no load on the base of the nut.

6.4.5 Maximum locking torque. The maximum locking torque is the highest self-locking torque encountered in any installation or removal cycle with no load on the base of the nut.

6.5 Subject term (key word) listing.
NUT
NICKEL ALLOY
SELF-LOCKING
220 KSI Ftu
900°F
INCONEL

Custodians:
Navy-AS
Air Force - 11
Army - AV

Reviewer Activity:
DLA - IS

Preparing activity:
Navy - AS
(Project No. 5310-1425)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 53
Lakehurst, NJ 08733-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 53
Lakehurst, NJ 08733-5100

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:27Z
Check the source to verify that this is the current version before use.